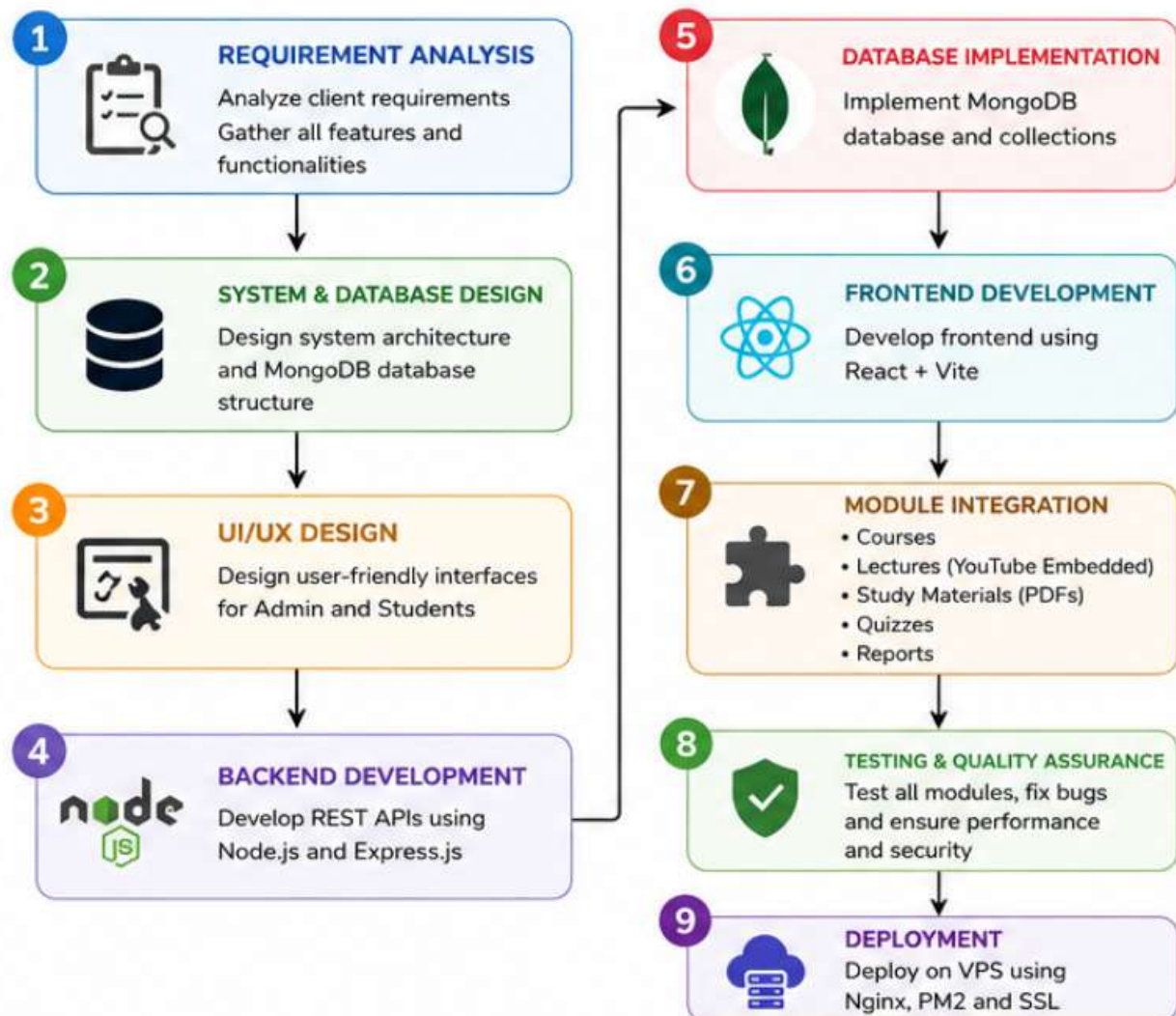


# METHODOLOGY DIAGRAM



## Phase 1: Requirement Analysis

First, we gather and analyze the client's requirements to understand the learning platform's objectives, target users, and required functionalities.

### Activities:

- Understand institute requirements
- Identify admin and student roles
- Define course structure
- Define quiz and learning material requirements
- Estimate expected student capacity

### Output:

- Software Requirement Specification (SRS)

## **Phase 2: System Design**

The system architecture, database structure, and workflow are designed to ensure scalability, security, and ease of use.

### **Activities:**

- Design LMS architecture
- Design MongoDB database
- Define user roles and permissions
- Create module structure

### **Output:**

- System Architecture Diagram
- Database Design

## **Phase 3: UI/UX Design**

User-friendly interfaces are designed for both administrators and students.

### **Activities:**

- Admin Dashboard Design
- Student Dashboard Design
- Course Page Design
- Quiz Interface Design
- Responsive Mobile Layout

### **Output:**

- Wireframes and User Interface Design

## **Phase 4: Development**

The LMS platform is developed using modern web technologies.

### **Frontend Technologies**

- React.js
- Vite
- Tailwind CSS

### **Backend Technologies**

- Node.js
- Express.js

### **Database**

- MongoDB

**Modules Developed**

- Authentication System
- Course Management
- Lecture Management
- Study Materials
- Quiz System

**Output:**

- Functional LMS Platform

**Phase 5: Testing**

The system is thoroughly tested to ensure reliability and performance.

**Activities:**

- Functional Testing
- User Authentication Testing
- Quiz Testing
- Performance Testing
- Security Testing

**Output:**

- Bug-Free System

**Phase 6: Deployment**

The completed LMS is deployed on the VPS server and made accessible online.

**Server Environment**

- Ubuntu VPS
- Nginx
- PM2
- MongoDB
- SSL Certificate

**Output:**

- Live LMS Website

**Phase 7: Training & Support**

The administrator is trained to manage courses, lectures, and quizzes effectively.

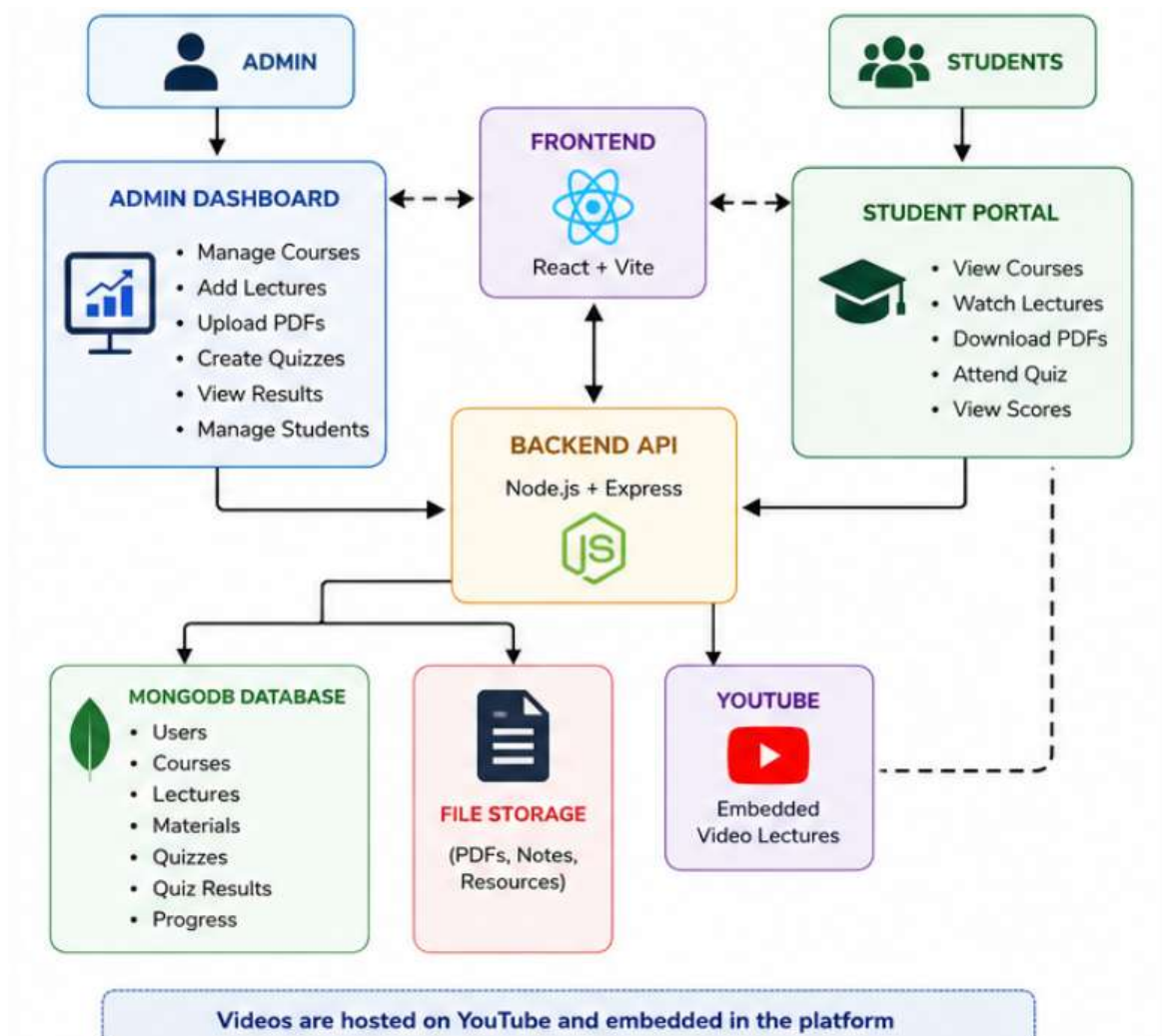
**Activities:**

- Admin Training
- Documentation
- Technical Support & Maintenance

**Output:**

- Fully Operational LMS Platform

# SYSTEM ARCHITECTURE



## Presentation Layer (Frontend)

This is the interface used by Admins and Students.

### Technology:

- React.js
- Vite
- Tailwind CSS

### Functions:

- Login & Registration
- Course Dashboard
- Watch Lectures
- Attend Quizzes
- Download Study Materials

## **Application Layer (Backend)**

This is the core business logic layer.

### **Technology:**

- Node.js
- Express.js

### **Functions:**

- User Authentication
- Course Management
- Quiz Evaluation
- Student Management
- API Handling
- Access Control

## **Data Layer (Database)**

All LMS data is stored here.

### **Technology:**

- MongoDB

### **Stores:**

- Users
- Courses
- Lectures
- PDFs
- Quizzes

# TECHNOLOGIES AND TOOLS USED

## Frontend Technologies

| Technology   | Purpose                    |
|--------------|----------------------------|
| React.js     | User Interface Development |
| Vite         | Fast Frontend Build Tool   |
| Tailwind CSS | Responsive UI Design       |
| Axios        | API Communication          |
| React Router | Navigation and Routing     |

## Backend Technologies

| Technology | Purpose                         |
|------------|---------------------------------|
| Node.js    | Server-Side Runtime Environment |
| Express.js | REST API Development            |
| JWT        | Authentication & Authorization  |
| bcrypt.js  | Password Encryption             |
| Multer     | File Upload Management          |

## Database

| Technology | Purpose                 |
|------------|-------------------------|
| MongoDB    | NoSQL Database          |
| Mongoose   | MongoDB Object Modeling |

## Security Technologies

| Technology                | Purpose                     |
|---------------------------|-----------------------------|
| JWT Authentication        | Secure Login                |
| bcrypt.js                 | Password Hashing            |
| Role-Based Access Control | Admin & Student Permissions |
| HTTPS/SSL                 | Secure Data Transmission    |